

End-Sem Syllabus:

Part 1: Data Preprocessing

- Data Types: Nominal, Ordinal, Interval, Ratio.
- Data Quality: Measurement, Noise, Precision, Bias, Accuracy
- Data Issues: Outliers, Missing, Inconsistent, Duplicate Data
- Data Reduction: Aggregation, Sampling, PCA, Curse of Dimensionality
- Feature Engineering: Selection methods, Extraction, Construction,
- Data Transformation: Discretization, Binarization, Normalization

Part 2: Similarity & Dissimilarity

- Euclidean Distance
- Manhattan Distance
- Similarity: SMC, Jaccard, Cosine
- Set Difference

Part 3: Classification

- Classification & Applications
- KNN: Weighted, Overfitting, Underfitting
- Cross-Validation
- Decision Trees: Elements, Attributes, Algorithm
- Splitting: Entropy, Gini
- Overfitting & Handling

Part 4: Association Rule Mining

- Association Rules: Support, Confidence, Lift
- Frequent Itemsets, Market Basket Analysis
- Apriori: Candidate Generation & Pruning, Rule Generation
- Closed & Maximal Frequent Sets
- Interestingness Measures & Limitations

Part 5: Clustering

- Types of clustering.
- K-Means, DBSCAN
- Silhouette Score and Elbow Method (only)

Part 6: Anomaly Detection

- Types & Applications
- Model-Based (3 sigma), Proximity-Based (KNN), Density & Clustering Approaches
- Evaluation (Confusion Matrix)